Gayatri Vidya Parishad College of Engineering Autonomous

Computer Science and Engineering | Course Code: CS201

Data Structures and Algorithms - Comprehensive Syllabus Lecture Notes

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Syllabus Focus: CS201 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Data Structures and Algorithms. Includes deta

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Asymptotic Analysis & Linear Structures

" Syllabus Scope: Big-O, Omega, Theta asymptotic bounds, Master Theorem, Amortized analysis, Stacks, Queu

Unit 2: Balanced Search Trees & Heaps

" Syllabus Scope: Binary Search Trees, AVL rotations, Red-Black tree coloring rules, Binary Heaps, Priori

CS201: Data Structures and Algorithms - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Graph Traversal & Minimum Spanning Trees

" Graph representations, BFS, DFS, Topological sorting, Prim and Kruskal Minimum Spanning Tree algorithms, Dis

Unit 4: Shortest Path Algorithms

" Dijkstra single-source shortest path with priority queues, Bellman-Ford algorithm for negative weights, Floy

Unit 5: Dynamic Programming & Greedy Design

" Optimal substructure, 0/1 Knapsack, Longest Common Subsequence LCS, Matrix Chain Multiplication, Huffman cod

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Solve the recurrence relation Tn = 3Tn/2 + On log n using the Master Theorem.
2. Insert the keys [15, 20, 24, 10, 13, 7, 30, 36] into an empty AVL tree and show all intermediate rotat
3. Formulate the dynamic programming recurrence and bottom-up matrix construction for the 0/1 Knapsack pr

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Thomas H. Cormen, Charles E. Leiserson - Introduction to Algorithms MIT Press 3rd Ed
* Mark Allen Weiss - Data Structures and Algorithm Analysis in C++ Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students